What does a quantum reservoir bring to physics-informed PDE solvers?

A fair benchmark on solid mechanics and phase-field equations

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Abstract

Quantum physics-informed neural networks (QPINNs) are often reported with few parameters and modest accuracy, but rarely against classical models of the same size tuned with the same care. We revisit the question with controlled comparisons on linear elasticity and on phase-field equations. Variational QPINNs are limited by optimization rather than expressivity: on the Allen–Cahn equation a 6-qubit QPINN reaches a relative error of 0.76. Freezing the circuit and solving only the linear readout by least squares (a quantum reservoir PINN, QERC-PINN) removes this obstacle and reaches $\sim 5 \times 10^{-5}$ on the Kirsch plate. We then prove that the reservoir cannot add representational power: its features span a subspace of the function space $\mathcal{F}_\rho$ fixed by the encoding, $\mathcal{F}_\rho$ is spanned by products of single-qubit functions, and classical product random features with $O(N)$ cost per feature give access to the same space; on a through-silicon via (TSV) they match the Haar reservoir (3.5 vs 3.9×10^{-3}). Reading the features out on hardware needs $O(N^2 2^N/\varepsilon^2)$ shots per collocation point, more than computing them classically. Across a Kirsch plate, TSVs in 2D, axisymmetry and 3D (with Abaqus references), and the Allen–Cahn and Cahn–Hilliard equations, least-squares solvers without any training beat published adaptive PINNs (Allen–Cahn 10^{-3} vs 2.3×10^{-2}; Cahn–Hilliard 6.9×10^{-4} vs 9.5×10^{-3}), but the quantum reservoir trails classical random features; on sharp phase-field interfaces it does not improve from 8 to 10 qubits (1.0×10^{-2}) while classical features reach 5×10^{-6}. Shot noise is the main obstacle on hardware; an energy formulation reduces it 3–4 times. We release all code and data.

1 Introduction

Physics-informed neural networks (PINNs) [1] solve PDEs by minimizing the equation residual over the parameters of a network. Their quantum counterparts (QPINNs) replace the network by a parametrized circuit [2, 3] and are often motivated by parameter efficiency and by the large function spaces that a few qubits can encode [4]. Such claims depend strongly on the classical baseline: a quantum model is typically compared with a PINN of a different size, trained differently, or not tuned.

We ask a narrow question with careful controls: *which part of a quantum PDE solver actually helps, and can a classical model with the same structure reproduce it?* We study linear elasticity (a plate with a hole, through-silicon vias in 2D, axisymmetry and 3D) and the phase-field equations of Allen–Cahn and Cahn–Hilliard type, whose sharp moving interfaces are a known difficulty for PINNs [5, 6]. Our quantum model is a quantum reservoir [7] with a least-squares readout, i.e. a physics-informed extreme learning machine [8] whose features are measurement probabilities of a fixed circuit; the closest classical analogues are random feature methods [9–12]. Quantum algorithms based on Carleman linearization address the same reaction–diffusion equations on fault-tolerant hardware [13, 14]; we target near-term devices instead.

Our contributions are:

Table 1: Cross-check of the reference solutions (relative L_2 error of displacement; von Mises as mean absolute error normalised by the maximum, as in Zhu et al. [15]).

Comparison	$\boldsymbol{u}$	von Mises, mid-plane	von Mises, near top
Axisymmetric, clamped: our FE vs Abaqus	2.2×10^{-4}	3.2×10^{-4}	6.5×10^{-4}
Axisymmetric, free top: our FE vs Abaqus	8.8×10^{-5}	3.2×10^{-4}	7.7×10^{-4}
3D clamped mid-plane vs 2D plane strain	2.4×10^{-4}	3.1×10^{-4}	–

1. A diagnosis of variational QPINNs: the bottleneck is optimization (Sec. 3); on the Allen–Cahn equation they fail where least-squares solvers succeed (Sec. 7).
2. Propositions showing that the reservoir only selects a subspace of an encoding-defined space that is classically accessible at $O(N)$ cost per feature, and that hardware readout is more expensive than classical evaluation (Sec. 4).
3. A fair benchmark with tuned classical random features and controls sharing the tensor-product structure, on solid mechanics (Secs. 5–6) and phase-field equations (Sec. 7), with independent references (analytic, our FE, Abaqus, spectral).
4. A shot-noise analysis and a first mitigation by an energy formulation (Sec. 8).

2 Methods

2.1 Problems

All problems are 2D linear elasticity in plane strain or plane stress: $\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}$, $\boldsymbol{\sigma} = \lambda \operatorname{tr}\boldsymbol{\varepsilon}\, \boldsymbol{I} + 2\mu\boldsymbol{\varepsilon} - \beta\Delta T\, \boldsymbol{I}$, $\beta = (3\lambda + 2\mu)\alpha$.

- **Kirsch plate**: an infinite plate with a circular hole under uniaxial tension, truncated at R, with the exact displacement on the outer boundary and traction-free hole ($K_t = 3$).
- **Single TSV**: a Cu via (radius a) in Si (outer radius $4a$), uniform cooling $\Delta T = -250\,\mathrm{K}$.
- **Periodic TSV array**: square array with pitch L; a quarter cell with symmetric roller conditions (zero normal displacement and zero shear traction) under zero macroscopic strain.
- **TSV unit cell in 3D and in axisymmetry**: the geometry of Zhu et al. [15] (via diameter 5 µm, height 50 µm, pitch 15 µm, $\Delta T = -250\,\mathrm{K}$, liner omitted). We solve the upper half with a symmetric mid-plane and either a clamped top (their scenario 1) or a traction-free top (the wafer surface). The axisymmetric model replaces the square cell by a cylinder of equal area; the 3D model is the quarter of the square cell.

References are analytic (Kirsch, single TSV) or a P1 finite-element solution on an interface-fitted mesh with $h = 0.005a$, checked against $2h$ (differences $\leq 3 \times 10^{-5}$ in displacement). For the 3D and axisymmetric cells we computed Abaqus references (CAX8, C3D10) and cross-checked them (Table 1): our axisymmetric FE and Abaqus agree to $< 10^{-3}$, the mid-plane stress at the via centre matches the plane-strain Lamé solution exactly ($\sigma_{rr} = \sigma_{\theta\theta} = 642\,\mathrm{MPa}$, $\sigma_{zz} = 917\,\mathrm{MPa}$), and the mid-plane of the clamped 3D cell coincides with the 2D plane-strain cell.

2.2 Least-squares PINN with fixed features

The displacement in each material phase is $\boldsymbol{u}(\boldsymbol{x}) = \sum_k \boldsymbol{c}_k \phi_k(\boldsymbol{x})$ with fixed features ϕ_k. PDE residuals at collocation points, boundary conditions and interface conditions (continuity of $\boldsymbol{u}$

and of traction, with the thermal-mismatch jump) are all linear in $\boldsymbol{c}$, so the coefficients follow from one regularized least-squares solve [8]. Each phase has its own features.

2.3 Features compared

- **QERC (quantum)**: per-qubit encoding $|\psi(\boldsymbol{x})\rangle = \bigotimes_{l=1}^{N} \left(\cos\frac{\theta_l}{2}|0\rangle + e^{i\varphi_l}\sin\frac{\theta_l}{2}|1\rangle\right)$, with $2N$ angles from a random affine map of the input (as in Ikeda et al. [7]), followed by a fixed reservoir U (Haar, Clifford+T or transverse-field Ising evolution); the 2^N outcome probabilities $p_k(\boldsymbol{x}) = |\langle k|U|\psi(\boldsymbol{x})\rangle|^2$ are the features.
- **tanh RF**: $\tanh(\boldsymbol{w}_k \cdot \boldsymbol{x} + b_k)$ [9].
- **Fourier RF / Fourier PIELM**: $\cos(\boldsymbol{w}_k \cdot \boldsymbol{x} + b_k)$; FS-PIELM [16].
- **Tensor random projection (classical control)**: $\mathrm{Re}\,\langle\psi(\boldsymbol{x})|R_k|\psi(\boldsymbol{x})\rangle$ with the same $|\psi(\boldsymbol{x})\rangle$ and independent GUE matrices R_k.
- **MLP2 (classical control)**: features of a random two-layer tanh network with $2N$ hidden units.

Every model has the same number of features (2^N) and its input scale is swept; for each seed we report the best scale, and we widen the sweep until the best scale lies inside it. This tuning was decisive: an untuned classical baseline looked equal to the quantum model with two seeds, while the tuned one was 20× better; and at $N = 12$ the cosine features improved from 3.2×10^{-4} to 8.3×10^{-6} once the sweep was extended beyond its original upper edge.

The quantum features are simulated exactly as state vectors in JAX, which provides the first and second derivatives with respect to the coordinates by automatic differentiation. We checked the implementation against Qiskit: for $N = 2$–6 the outcome probabilities agree to 4×10^{-16} (Statevector), and with 2×10^5 shots (Aer, $N = 4$) the sampled frequencies agree within 2.1 standard errors.

3 Variational QPINNs are limited by optimization

On the 1D Lamé problem, a supervised fit of the analytic solution reaches 7×10^{-4} while PDE training of the same circuit stops at 5×10^{-2}. On the Kirsch plate, polar or Chebyshev encodings improve the displacement error by 2–3 times, but a classical MLP of the same size with the same inputs stays ahead. Since the analytic solution lies close to the circuit's function class, we attribute the gap to optimization.

4 What the reservoir can and cannot add

Write the per-qubit encoding as $\rho(\boldsymbol{x}) = \bigotimes_{l=1}^{N} \rho_l(\boldsymbol{x})$, $\rho_l = |q_l\rangle\langle q_l|$, $|q_l\rangle = \cos\frac{\theta_l}{2}|0\rangle + e^{i\theta_{N+l}}\sin\frac{\theta_l}{2}|1\rangle$, and let

$$\mathcal{F}_\rho = \left\{\boldsymbol{x} \mapsto \mathrm{Tr}[H\rho(\boldsymbol{x})] \;:\; H = H^\dagger \in \mathbb{C}^{2^N \times 2^N}\right\} \tag{1}$$

be the real function space spanned by the matrix elements of $\rho(\boldsymbol{x})$.

Proposition 1 (the reservoir selects a subspace)**.** *For any unitary U, the features $p_k(\boldsymbol{x}) = |\langle k|U|\psi(\boldsymbol{x})\rangle|^2$ satisfy* $\mathrm{span}\{p_k\} = \{\mathrm{Tr}[H\rho] : H \in \mathcal{D}_U\} \subseteq \mathcal{F}_\rho$*, where $\mathcal{D}_U = \mathrm{span}_{\mathbb{R}}\{U^\dagger|k\rangle\langle k|U\}$ is a commutative algebra of dimension 2^N. Hence* $\dim \mathrm{span}\{p_k\} \le \min(2^N, \dim \mathcal{F}_\rho)$*, and $\sum_k p_k \equiv 1$.*

Proof. $p_k = \mathrm{Tr}[M_k\rho]$ with $M_k = U^\dagger|k\rangle\langle k|U$ and the trace is linear in M_k; the M_k are orthogonal projectors summing to I. □

Proposition 2 (product structure of $\mathcal{F}_\rho$)**.** $\mathcal{F}_\rho = \operatorname{span}\{\prod_{l=1}^{N} \operatorname{Tr}[A_l \rho_l(\boldsymbol{x})] \ : \ A_l = A_l^\dagger \in \mathbb{C}^{2\times 2}\}$, *and each factor lies in* $\operatorname{span}\{1,\ \boldsymbol{n}_l(\boldsymbol{x})\}$*, where* $\boldsymbol{n}_l = (\sin\theta_l \cos\theta_{N+l}, \sin\theta_l \sin\theta_{N+l}, \cos\theta_l)$ *is the Bloch vector of qubit* l*. In particular* $\dim \mathcal{F}_\rho \le 4^N$*.*

Proof. Product operators $\bigotimes_l A_l$ span the Hermitian matrices over $\mathbb{R}$, and $\operatorname{Tr}[(\bigotimes_l A_l)\bigotimes_l \rho_l] = \prod_l \operatorname{Tr}[A_l \rho_l]$. Writing $A_l = a_0 I + \boldsymbol{a}\cdot\boldsymbol{\sigma}$ gives $\operatorname{Tr}[A_l\rho_l] = a_0 + \boldsymbol{a}\cdot\boldsymbol{n}_l$. □

Corollary 1 (classical access to the same space)**.** *The product random features* $f_k(\boldsymbol{x}) = \prod_l \operatorname{Tr}[A_{k,l}\rho_l(\boldsymbol{x})]$ *with random Hermitian* $A_{k,l}$ *lie in* $\mathcal{F}_\rho$ *and cost* $O(N)$ *operations each, without a* 2^N*-dimensional state. The tensor random projection of Sec. 6 (*$\operatorname{Re}\langle\psi|R_k|\psi\rangle$*, GUE* R_k*) also lies in* $\mathcal{F}_\rho$ *but costs* $O(4^N)$ *per feature.*

Proposition 3 (readout cost on hardware)**.** *Let each* p_k *be estimated from* S *shots. Then* $\operatorname{Var}\hat{p}_k = p_k(1-p_k)/S$*; for a Haar reservoir* $\mathbb{E}\, p_k = 2^{-N}$*, so the relative standard deviation is* $\approx \sqrt{2^N/S}$ *and a relative accuracy* ε *on the features needs* $S \gtrsim 2^N/\varepsilon^2$ *shots per circuit. The second derivative* $\partial^2 p_k/\partial x^2 = \sum_{ij} \partial_{\theta_i\theta_j} p_k\, \theta_i'\theta_j' + \sum_i \partial_{\theta_i} p_k\, \theta_i''$ *requires* $O(N^2)$ *parameter-shifted circuits, so one collocation point costs* $O(N^2\, 2^N/\varepsilon^2)$ *shots.*

Proof. Multinomial sampling; the parameter-shift rule [17, 18] gives each $\partial_{\theta_i\theta_j} p_k$ from four shifted circuits, and there are $(2N)^2$ pairs. □

Corollary 2 (no advantage from feature evaluation)**.** *For* $m = 2^N$ *features at* M *points, the quantum readout costs* $O(MN^2 2^N/\varepsilon^2)$ *shots, while the classical product features of Corollary 1 cost* $O(MN2^N)$ *operations, with the same function space* $\mathcal{F}_\rho$ *available to both. The ratio is* $N/\varepsilon^2 \ge 1$ *for any useful* $\varepsilon < 1$*.*

Together, Propositions 1–3 say that (i) what a quantum reservoir PINN can represent is fixed by the encoding, (ii) the same space is classically accessible at linear cost in N, and (iii) reading it out on hardware is more expensive than computing it classically. A benefit of the quantum device can therefore only come from the particular 2^N-dimensional subspace that a reservoir selects; Sections 5–6 test whether that subspace is better than random ones. On the single TSV ($N = 8$, 3 seeds), the product random features of Corollary 1 reach 3.5×10^{-3} against 3.9×10^{-3} for the Haar reservoir, with the same best input scale and the same per-seed ordering ($5.1/1.6/5.5\times10^{-3}$ versus $5.8/1.8/5.6\times10^{-3}$): the classical $O(N)$-cost features are at least as good.

Remark. Proposition 3 concerns outcome probabilities. Observables with $O(1)$ expectation values (e.g. a few Pauli strings) avoid the 2^N factor in S but give fewer features; this is left for future work.

Two numerical consequences are confirmed in Fig. 1: (i) if every qubit receives the same input, $\dim \mathcal{F}_\rho$ is small and the reservoir does not matter (changing the Ising time or couplings leaves the PDE error unchanged to three digits; at $N = 8$ only 152 of 256 features are independent); (ii) with a per-qubit encoding, $\dim \mathcal{F}_\rho$ grows like 4^N and a generic reservoir (Clifford+T, Ising, Haar) improves the error by two orders of magnitude over no reservoir or a Clifford-only reservoir (as observed for classification by Ikeda et al. [7]): a reservoir that mixes the computational basis is needed to select a rich subspace, but any generic one does.

5 Fair comparison on the Kirsch plate

With tuned scales on both sides (Table 2, Fig. 2), the quantum reservoir and tanh random features are equivalent, and both recover $K_t = 3.00 \pm 0.03$. Cosine features — the classical analogue closest to the trigonometric quantum features — are about $40\times$ more accurate.

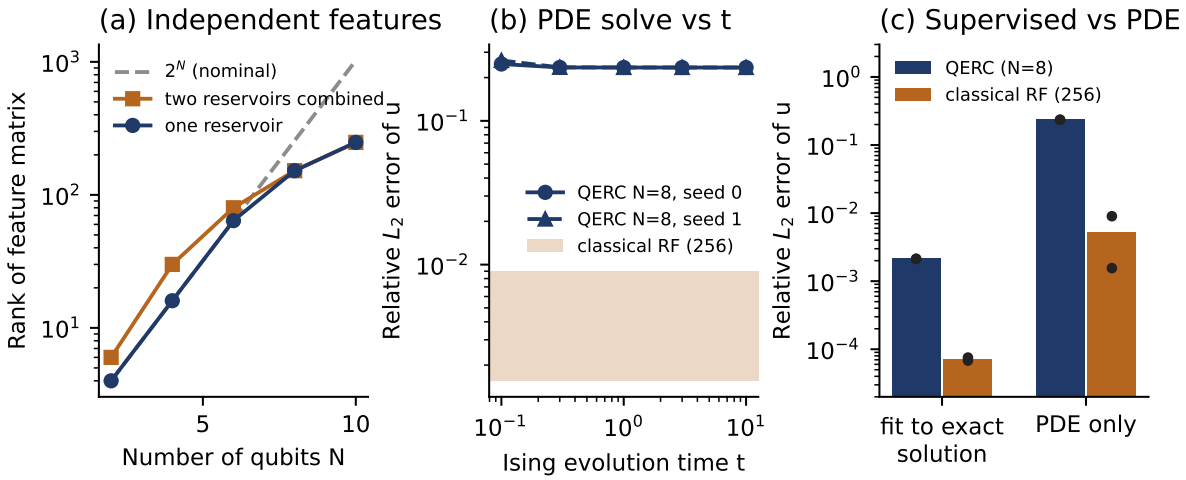


Figure 1: Reservoir and function space (Kirsch, PDE only). (a) Rank of the feature matrix versus the number of qubits for one reservoir and for two reservoirs combined, with the nominal 2^N. (b) PDE error versus the Ising evolution time ($N = 8$, two seeds) and the range of classical random features with 256 features. (c) Supervised fit of the exact solution versus the PDE-only solve, QERC ($N = 8$) and classical random features.

Table 2: Kirsch plate, PDE only, $N = 10$ (1024 features), 5 seeds, best input scale per seed. Relative L_2 error of displacement: geometric mean [min, max].

Features	rel. $L_2(\boldsymbol{u})$
QERC, Haar	4.6×10^{-5} [2.9×10^{-5}, 8.4×10^{-5}]
QERC, Clifford+T	4.7×10^{-5} [2.4×10^{-5}, 9.7×10^{-5}]
tanh RF	5.9×10^{-5} [3.2×10^{-5}, 1.0×10^{-4}]
Fourier PIELM	$\mathbf{1.1 \times 10^{-6}}$ [4.8×10^{-7}, 1.9×10^{-6}]
FS-PIELM	1.3×10^{-6} [4.4×10^{-7}, 6.2×10^{-6}]

6 Problems with material interfaces

6.1 Single TSV

For a single Cu via in Si ($N = 8$, 5 seeds), QERC reaches 4.6×10^{-3}, ten times better than one-layer tanh (4.8×10^{-2}) or Fourier (4.3×10^{-2}) features. The tensor random projection reproduces the quantum result (4.4×10^{-3}, with matching per-seed and per-scale trends; Fig. 3), and a random two-layer network reaches 1.2×10^{-2}. The advantage over one-layer features therefore comes from the product structure of $\mathcal{F}_\rho$, not from anything specific to quantum measurement. The advantage also disappears with more features: at $N = 10$ QERC gives 1.5×10^{-4} (tanh 2.1×10^{-3}, Fourier 1.5×10^{-3}), and at $N = 12$ the cosine features with a properly extended scale sweep give 8.3×10^{-6}, about 7× better than QERC (5.6×10^{-5}) and tanh (5.7×10^{-5}).

6.2 Periodic TSV array

For the quarter unit cell (Fig. 4), cosine features are best for narrow pitches ($L = 3, 4$: 1.4×10^{-5} and 2.7×10^{-5}, close to the FE reference accuracy). For wider pitches the cosine features degrade fastest ($L = 10$: 4.4×10^{-3} vs 1.0×10^{-3} for QERC). This is not a collocation-density effect (doubling the points changes nothing) but a feature-count effect: with 1024 features at $L = 6$, all models drop to $\sim 10^{-5}$ and the cosine features are again best (5.9×10^{-6}). Across all pitches, QERC and MLP2 agree within the seed spread.

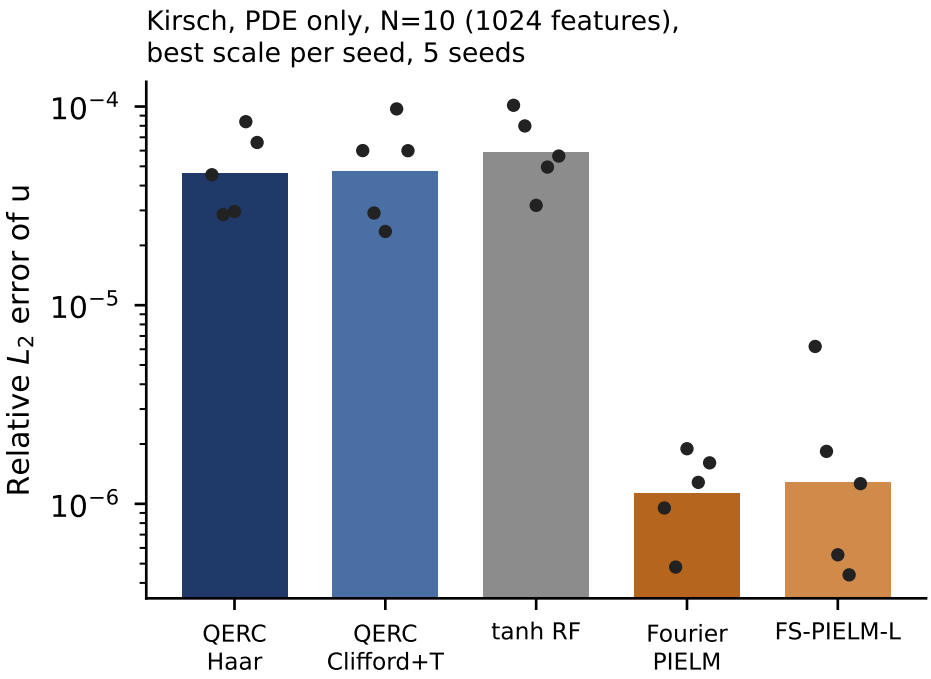


Figure 2: Kirsch plate: all feature types with tuned input scale (5 seeds).

Table 3: Axisymmetric TSV cell, $N = 8$, 5 seeds: relative $L_2(\boldsymbol{u})$ and normalised von Mises error near the top.

	clamped top		free top	
Features	$\boldsymbol{u}$	von Mises, top	$\boldsymbol{u}$	von Mises, top
QERC, Haar	2.7×10^{-2}	9.8×10^{-3}	9.9×10^{-3}	2.0×10^{-2}
MLP2	2.7×10^{-2}	8.8×10^{-3}	8.1×10^{-3}	2.1×10^{-2}
tanh RF	1.5×10^{-2}	8.0×10^{-3}	3.2×10^{-3}	1.5×10^{-2}
Fourier RF	$\mathbf{6.0 \times 10^{-3}}$	$\mathbf{4.7 \times 10^{-3}}$	$\mathbf{1.5 \times 10^{-3}}$	$\mathbf{6.5 \times 10^{-3}}$

6.3 Axisymmetric and 3D unit cells: the near-surface region

The region that matters for device placement is just below the wafer surface, which a 2D plane-strain model cannot represent. In the axisymmetric cell ($N = 8$, 5 seeds; Table 3) the near-surface von Mises error is 3–15 times the mid-plane error for every model, and the cosine features are the most accurate in both the clamped and the free-top case; QERC is on par with the random two-layer network. The 3D cell with the Abaqus reference shows the same ordering at $N = 10$ (clamped: QERC 2.9×10^{-2} vs Fourier 1.3×10^{-2} in $\boldsymbol{u}$; free top: 8.0×10^{-3} vs 4.4×10^{-3}).

6.4 A high-dimensional counterexample

One may expect the exponential tensor-product space to help when the input is high-dimensional. For the dispersion relation of a d-layer laminate ($d = N$ inputs), whose exact solution lies in the 2^N-dimensional span of $\bigotimes_i(\cos\varphi_i, \sin\varphi_i)$, all random feature models fail (relative error > 1), quantum or classical: a random reservoir picks a random 2^N-dimensional subspace of a 3^N–4^N-dimensional space, whose overlap with the target shrinks exponentially. A benefit would require a reservoir designed for the problem.

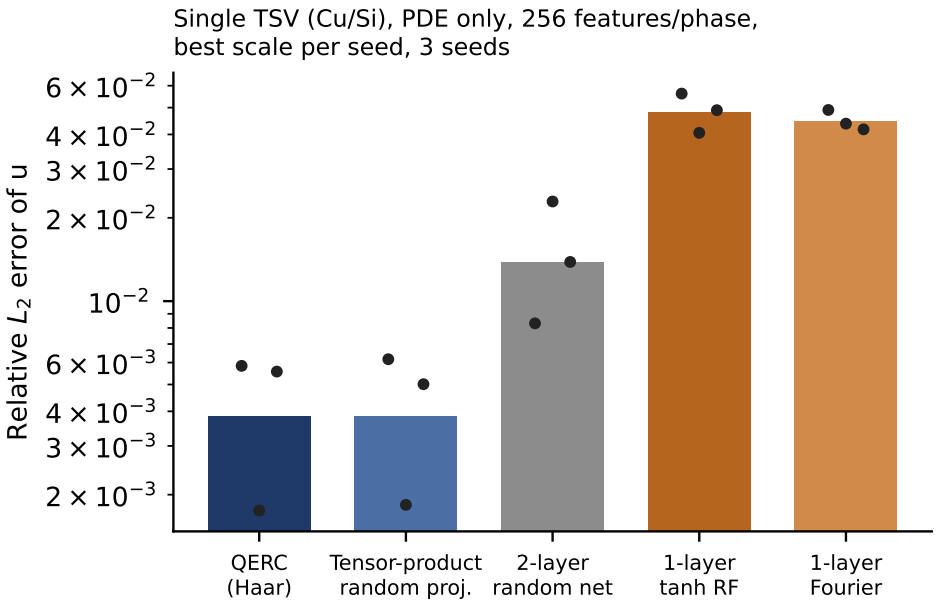


Figure 3: Single TSV: quantum reservoir and classical controls.

7 Phase-field equations

7.1 Problems and solver

We take the benchmarks of Raissi et al. [1] and Wight and Zhao [5], $x \in [-1, 1]$, $t \in [0, 1]$, periodic:

$$\text{Allen–Cahn:} \quad u_t - 10^{-4} u_{xx} + 5u^3 - 5u = 0, \qquad u(0, x) = x^2 \cos \pi x, \tag{2}$$

$$\text{Cahn–Hilliard:} \quad u_t = \mu_{xx}, \quad \mu = \gamma_2(u^3 - u) - \gamma_1 u_{xx}, \qquad \gamma_1 = 10^{-6},\ \gamma_2 = 10^{-2}, \quad u(0, x) = -\cos 2\pi x, \tag{3}$$

plus a smooth Cahn–Hilliard control ($\gamma_2 = 1$, $\gamma_1 = 0.02$). Both are gradient flows of a Ginzburg–Landau energy; the Cahn–Hilliard flow also conserves $\int u \, \mathrm{d}x$. The interfaces are about 0.04 (Allen–Cahn at $t = 1$) and 0.01 (Cahn–Hilliard) wide. The solution is expanded in fixed features of $\boldsymbol{z}(x) = (\cos \pi x, \sin \pi x)$, which makes it periodic by construction. We march in time with BDF2 ($\Delta t = 10^{-3}$), linearize u^3 by Newton's method and solve one least-squares problem per Newton iteration; Cahn–Hilliard is solved in mixed form, so only second derivatives occur and u and μ are found together. The feature matrices are built once and reused at every step. References are Fourier pseudo-spectral solutions with ETDRK4 ($\Delta t = 10^{-5}$); with an exact Fourier basis our time stepping reproduces them to 4.9×10^{-6} (Allen–Cahn) and 6×10^{-10} (smooth Cahn–Hilliard). For comparison we train a variational QPINN (6 qubits, 6 data re-uploading layers, 223 parameters) and classical PINNs (a 4×64 tanh MLP and one with 217 parameters) on the same space–time residual and initial-condition loss with Adam (2×10^4 steps).

7.2 Results

Table 4 shows three things. First, the least-squares solvers, quantum or classical, are far more accurate than the trained models: the variational QPINN does not capture the dynamics, our plain PINN reaches 0.13, and even the adaptive PINN of Wight and Zhao [5] is worse than every least-squares model. Second, the classical random features improve by two to three orders of magnitude from 256 to 1024 features, while the quantum reservoir stays at 10^{-2}. Proposition 2 explains why more qubits need not help: $\mathcal{F}_\rho$ consists of products of first-degree trigonometric

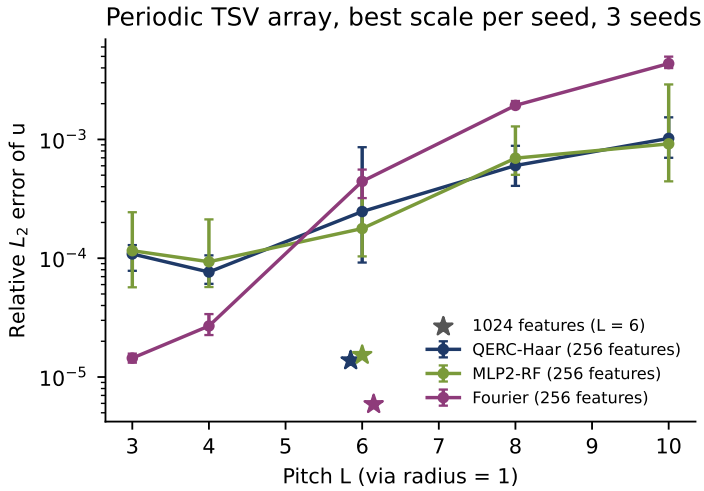


Figure 4: Periodic TSV array: displacement error versus pitch L (256 features; stars: 1024 features at $L = 6$).

Table 4: Allen–Cahn: relative L_2 error of u at $t = 1$. Least-squares models: geometric mean of the best scale per seed (5 seeds at $N = 8$, 3 at $N = 10$). Trained models: 3 seeds.

Model	$N = 8$ (256 features)	$N = 10$ (1024 features)
QERC, Haar (least squares)	1.0×10^{-2}	9.9×10^{-3}
tanh RF (least squares)	1.0×10^{-3}	$\mathbf{5.3 \times 10^{-6}}$
Fourier RF (least squares)	2.3×10^{-3}	3.4×10^{-5}
Variational QPINN (Adam)	0.76	–
PINN, 4×64 (Adam)	0.13	–
Adaptive PINN [5]	2.33×10^{-2}	

functions of bounded angles, whose spatial resolution is set by the input scale rather than by N, and a sharp interface needs fine resolution. Third, every model preserves energy dissipation: the discrete free energy decreases at every step ($\max_n(E^{n+1} - E^n) \leq 0$ to round-off). For the tanh and Fourier features at $N = 10$ the best scale lies at the upper edge of the sweep, so their errors are upper bounds.

On the sharp Cahn–Hilliard benchmark ($N = 8$, 5 seeds), tanh and Fourier features reach 6.9×10^{-4} and 1.4×10^{-3}, more than ten times better than the best adaptive PINN of Wight and Zhao [5] (9.5×10^{-3}); the quantum reservoir reaches 5.0×10^{-2}. Mass is conserved to 7×10^{-6} (classical) and 3×10^{-4} (quantum). The smooth control is solved to 10^{-9}–10^{-10} by all models and does not discriminate between them.

Because the feature matrices do not depend on the coefficients or the initial condition, one quantum feature set solves families of problems: with the same $N = 8$ reservoir we solved nine Allen–Cahn problems (three initial conditions, $D \in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$) with errors from 9×10^{-4} to 10^{-1}, the error growing as the interfaces sharpen. On hardware the circuits would be measured once for all of them, whereas a variational QPINN is retrained for each problem.

8 Shot noise and an energy formulation

On hardware, p_k and its derivatives are estimated from S shots per circuit, with derivatives from the parameter-shift rule [17, 18]. We propagate the multinomial variance $p(1 - p)/S$ through the chain rule (an approximate model that ignores correlations between circuits) and add the

Table 5: Kirsch, $N = 10$, Haar reservoir, 2 seeds: relative $L_2(\boldsymbol{u})$ versus shots per circuit. Ridge and boundary penalty chosen by the test error for all methods (optimistic bound).

Shots	Strong form	Strong, split shots	Energy (Ritz)	Energy, split shots
10^3	0.46	0.72	**0.32**	0.66
10^5	0.25	0.25	**0.083**	0.16
10^7	0.062	0.13	**0.014**	0.035
∞	6.3×10^{-4}	–	$\mathbf{2.1 \times 10^{-4}}$	–

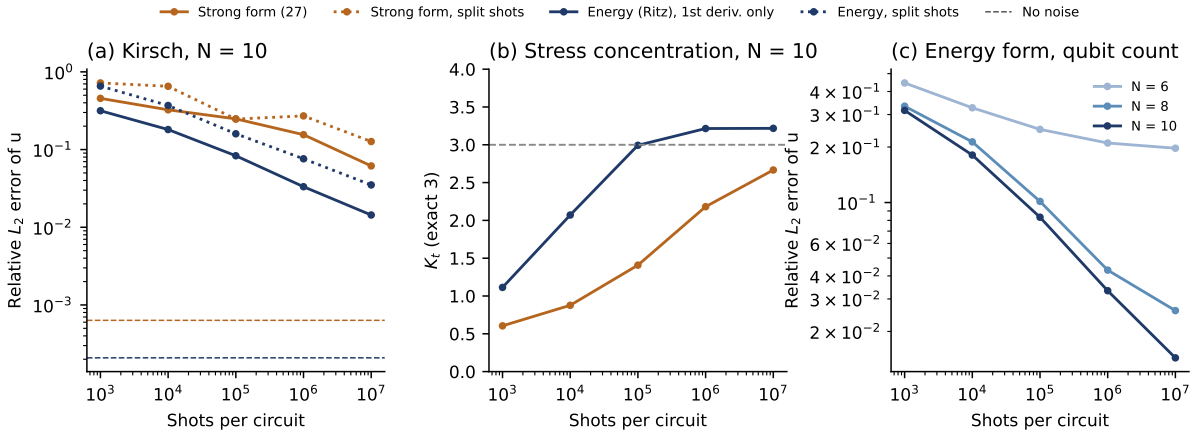


Figure 5: Shot noise: strong versus energy formulation, with and without split-shot estimation, and qubit count.

noise only when assembling the system. Since $p_k \approx 2^{-N}$, the relative noise per feature scales as $\sqrt{2^N/S}$, and second derivatives amplify it.

The energy (Ritz) formulation [19] minimizes $\frac{1}{2}\int \boldsymbol{\varepsilon} : \mathbb{C} : \boldsymbol{\varepsilon}\, dA$ plus a boundary penalty, so the stiffness matrix needs only first derivatives; the traction-free hole is a natural boundary condition. It reduces the error 3–4 times at every shot level and recovers $K_t = 3.0$ from 10^5 shots, where the strong form gives 1.4 (Table 5, Fig. 5). Splitting the shots into two independent halves removes the noise-squared bias of the normal matrix but makes the results worse: the bias acts as a ridge regularizer. Fewer qubits reduce the relative noise but lose expressivity ($N = 6$ saturates at 0.2). Even so, 10^7 shots per circuit still give 1.4% error, 70 times the noiseless value.

9 Discussion

Where the accuracy comes from. The large gap between trained and least-squares models (Table 4) is not quantum: it comes from replacing a non-convex optimization by a linear solve, as in classical extreme learning machines and random feature methods [8, 10, 11]. A quantum circuit used as a fixed feature map inherits this benefit, but so does any classical feature map.

What the quantum part can add. By Propositions 1–3, the representable space is set by the encoding, is classically accessible at $O(N)$ cost per feature, and is more expensive to read out on hardware than to compute. Numerically, classical controls with the same tensor-product structure reproduce the reservoir (TSV), and simple random features beat it on sharp interfaces by two to three orders of magnitude at $N = 10$. A quantum benefit would need a reservoir designed for the problem, or a readout whose cost does not grow as 2^N.

Fair baselines. Tuning changes conclusions by more than an order of magnitude: an untuned classical baseline looked equal to the quantum model, and widening the scale sweep improved the cosine features at $N = 12$ by a factor of 40. We recommend reporting the scale sweep and checking that the selected value is interior.

Limitations. All quantum results are noiseless state-vector simulations except for the shot-noise model of Sec. 8; the phase-field problems are one-dimensional; and the reservoirs are generic (Haar, Clifford+T, Ising) rather than designed. Two-dimensional phase-field problems and readouts with $O(1)$ expectation values are the natural next steps.

10 Conclusion

A quantum reservoir with a least-squares readout is a working PDE solver: it avoids the optimization failure of variational QPINNs and, on phase-field equations, outperforms trained PINNs. Under a fair comparison, however, its accuracy is matched or exceeded by classical random features with the same least-squares solver, its function space is classically accessible at linear cost in the number of qubits, and its hardware readout is more expensive than classical evaluation. We hope the benchmark, the controls and the propositions serve as a baseline for future claims of quantum advantage in physics-informed learning.

Code and data. `https://github.com/mmc-research-group/nishioka-qpinn-solid`

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